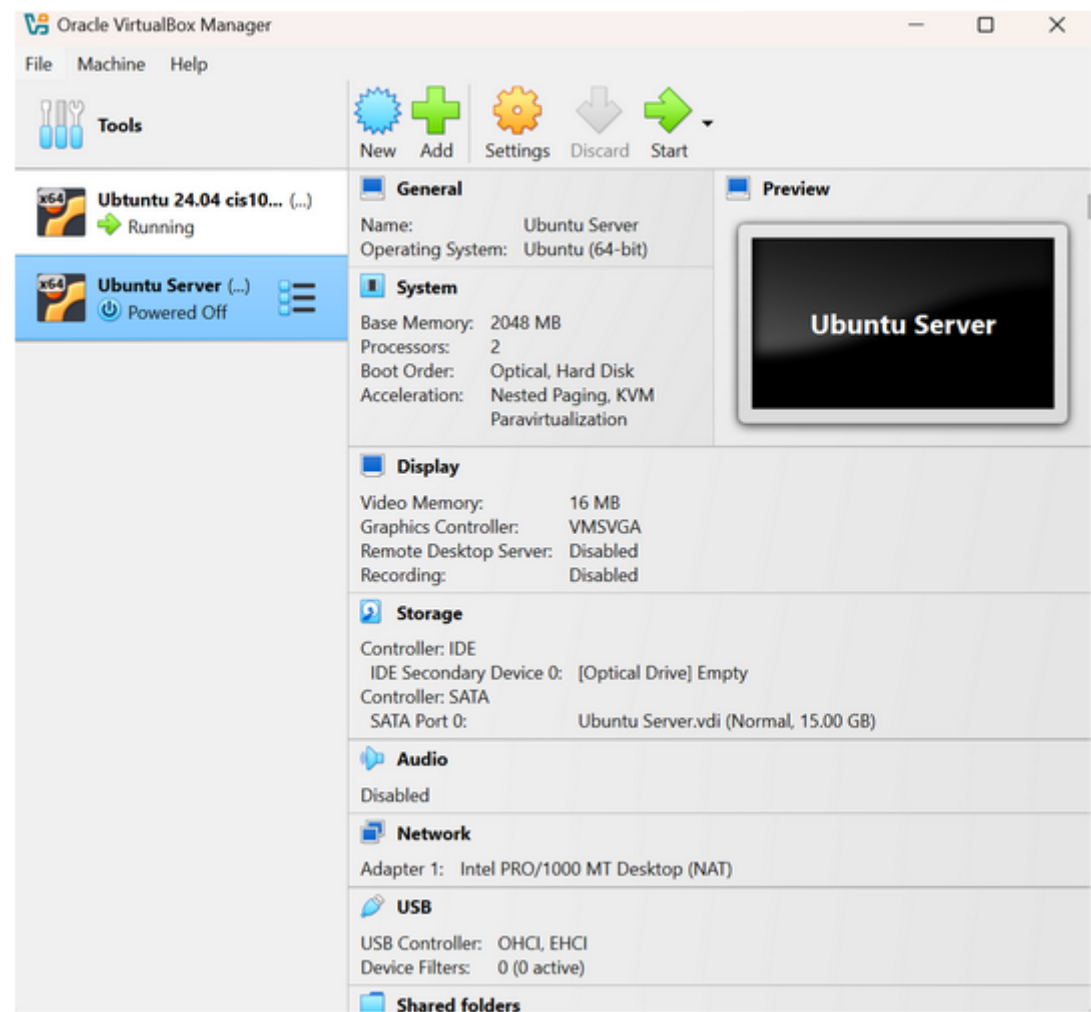
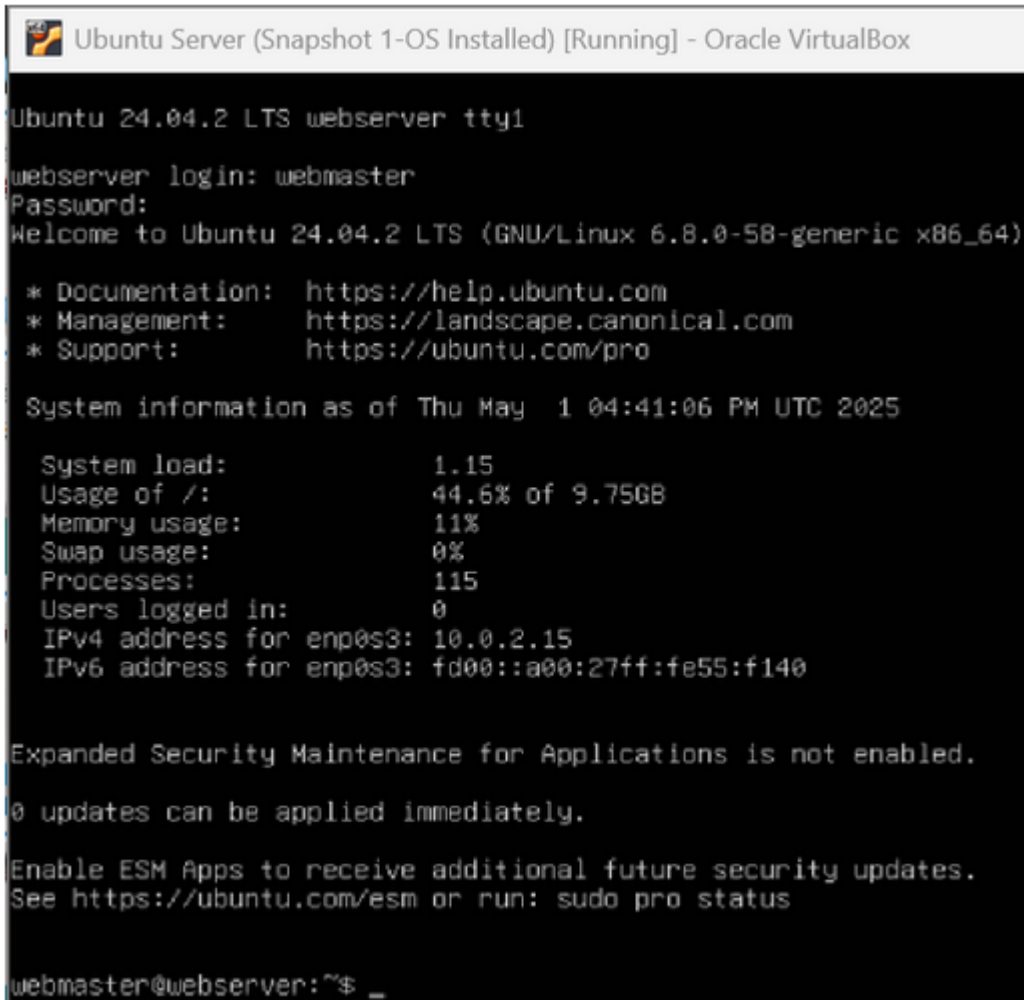


Deliverable 2 Submission

Server Specifications



Ubuntu Login Screen



```
Ubuntu Server (Snapshot 1-OS Installed) [Running] - Oracle VirtualBox
Ubuntu 24.04.2 LTS webserver tty1
webserver login: webmaster
Password:
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-58-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu May  1 04:41:06 PM UTC 2025

System load:          1.15
Usage of /:            44.6% of 9.75GB
Memory usage:         11%
Swap usage:           0%
Processes:            115
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd00::a00:27ff:fe55:f140

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

webmaster@webserver:~$
```

Questions

3. What is the IP address of your Ubuntu Server Virtual Machine? IPv4 Address: 10.0.2.15
4. How do you enable the Ubuntu Firewall? `sudo ufw enable`
5. How do you check if the Ubuntu Firewall is running? `sudo ufw status`
6. How do you disable the Ubuntu Firewall? `sudo ufw disable`
7. How do you add Apache to the Firewall? `sudo ufw allow 'Apache'`
8. What is the command you used to install Apache? `sudo apt update`
`sudo apt install apache2 -y`
9. What is the command you use to check if Apache is running? `sudo systemctl status apache2`
10. What is the command you use to stop Apache? `sudo systemctl stop apache2`
11. What is the command you use to restart Apache? `sudo systemctl restart apache2`
12. What is the command used to test Apache configuration? `sudo apache2ctl configtest`
13. What is the command used to check the installed version of Apache? `apache2 -v`
14. What are the most common commands to troubleshoot Apache errors? Provide a brief description of each command.

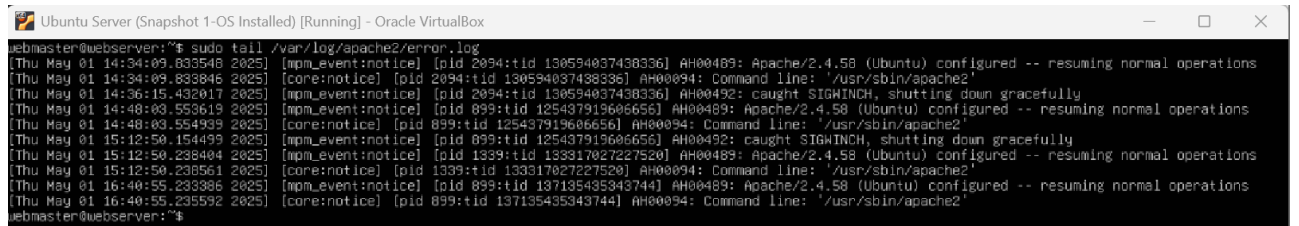
- `sudo ufw status` Ensure Apache traffic isn't being blocked
- `sudo tail -f /var/log/apache2/error.log` Live view of Apache error log

1. Which are Apache Log Files, and what are they used for? Provide examples and screenshots. this is a Apache Log /var/log/apache2/

- `access.log` All incoming HTTP requests (who accessed what and when)

```
webmaster@webserver:~$ sudo tail /var/log/apache2/access.log
[sudo] password for webmaster:
10.0.2.2 - - [01/May/2025:14:56:00 +0000] "GET / HTTP/1.1" 200 3460 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36 Edg/135.0.0.0"
10.0.2.2 - - [01/May/2025:14:56:00 +0000] "GET /icons/ubuntu-logo.png HTTP/1.1" 200 3607 "http://127.0.0.1:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36 Edg/135.0.0.0"
10.0.2.2 - - [01/May/2025:14:56:00 +0000] "GET /favicon.ico HTTP/1.1" 404 489 "http://127.0.0.1:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36 Edg/135.0.0.0"
10.0.2.2 - - [01/May/2025:14:56:52 +0000] "-" 408 0 "-" "-"
10.0.2.2 - - [01/May/2025:15:14:10 +0000] "GET / HTTP/1.1" 200 450 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36 Edg/135.0.0.0"
10.0.2.2 - - [01/May/2025:17:01:04 +0000] "GET /favicon.ico HTTP/1.1" 404 490 "http://127.0.0.1:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36 Edg/135.0.0.0"
webmaster@webserver:~$
```

- `sudo tail /var/log/apache2/error.log` All errors Apache encounters (useful for debugging)



```
webmaster@webserver:~$ sudo tail /var/log/apache2/error.log
[Thu May 01 14:34:09.833548 2025] [mpm_event:notice] [pid 2894:tid 130594037438336] AH00489: Apache/2.4.58 (Ubuntu) configured -- resuming normal operations
[Thu May 01 14:34:09.833846 2025] [core:notice] [pid 2894:tid 130594037438336] AH00492: caught SIGWINCH, shutting down gracefully
[Thu May 01 14:36:15.432017 2025] [mpm_event:notice] [pid 2894:tid 130594037438336] AH00492: caught SIGWINCH, shutting down gracefully
[Thu May 01 14:40:03.553619 2025] [mpm_event:notice] [pid 899:tid 125437919606656] AH00489: Apache/2.4.58 (Ubuntu) configured -- resuming normal operations
[Thu May 01 14:40:03.554939 2025] [core:notice] [pid 899:tid 125437919606656] AH00492: caught SIGWINCH, shutting down gracefully
[Thu May 01 15:12:50.238404 2025] [mpm_event:notice] [pid 1339:tid 133317027227520] AH00489: Apache/2.4.58 (Ubuntu) configured -- resuming normal operations
[Thu May 01 15:12:50.238561 2025] [core:notice] [pid 1339:tid 133317027227520] AH00492: caught SIGWINCH, shutting down gracefully
[Thu May 01 16:40:55.233386 2025] [mpm_event:notice] [pid 899:tid 137135435343744] AH00489: Apache/2.4.58 (Ubuntu) configured -- resuming normal operations
[Thu May 01 16:40:55.235592 2025] [core:notice] [pid 899:tid 137135435343744] AH00492: caught SIGWINCH, shutting down gracefully
webmaster@webserver:~$
```